Viraj Dharod

Experimental Quantum Physicist

+1 (805) 637-6325
dharodviraj@gmail.com
LinkedIn Google Scholar

Summary

- Experimental physicist with 5+ years of experience in quantum hardware engineering and applied device physics.
- Built multiple AMO, quantum optics, and fiber photonics experiments from the ground up, carrying them from measurement concept through hardware integration, characterization, and iterative optimization, producing validated results that led to peer-reviewed publications.
- Extensive Python experience for experimental control, data acquisition, parameter extraction, modeling, and analysis of large and noisy experimental datasets.

Research & Professional Experience

Graduate Student Researcher • University of California, Santa Barbara

Advisor: Prof. Ania C. Bleszynski Jayich Oct 2020 - Present [expected Dec 2026 completion]

- **Optical & photonics systems:** Designed, built, and optimized fiber and free-space optical experiments integrating lasers, photodetectors, AOMs, EOMs, RF electronics, and synchronized control hardware for laser-cavity locking, optical power stabilization, and pulsed optical measurements. Developed measurement procedures, characterized system performance, and troubleshot optical and electronic limitations. Mentored undergraduates and junior graduate students in setup, alignment, and troubleshooting.
- **Experimental design, characterization, and modeling:** Designed and executed cryogenic experiments to characterize diamond optomechanical devices, demonstrating state-of-the-art device performance by measuring mechanical quality factors greater than 10 million. Identified performance-limiting mechanisms, including optical-absorption-induced heating, and developed quantitative models to interpret experimental results and guide device optimization.
- **Cryogenics & quantum ground-state operation:** Built out and commissioned a dilution-refrigerator measurement platform, integrating cryogenic optical, RF, DC, and positioning hardware for device measurements below 250 mK. Demonstrated operation of GHz-frequency diamond optomechanical devices in the quantum ground state of mechanical motion using direct thermometry.
- **Python framework & automation:** Developed a modular Python control and data-acquisition framework integrating lasers, RF/microwave electronics, DAQs, cryogenic instrumentation, and optical hardware, enabling automated characterization, closed-loop control, synchronized measurements, and parameter extraction.
- **Collaborative research & experimental strategy:** Co-developed and published experimental protocols for advanced quantum sensing and readout, collaborating with theorists and experimentalists across UCSB, Stanford, and the University of Chicago to translate theoretical proposals into experimentally implementable approaches.

Undergraduate Research Scholar • Research School of Physics, Australian National University, Canberra

Advisor: Prof. Marcus W. Doherty May 2019 - August 2019

- **Hardware interfacing & software control:** Designed and built hardware and software to control a custom 3D vector electromagnet aligned coaxially with a confocal microscope for magnetometry experiments.
- **NV-center quantum microscopy & magnetometry:** Developed and implemented a discrete 3D magnetic field vector sweep protocol to determine the crystallographic orientation of individual NV centers based on optically detected magnetic resonance (ODMR) spectroscopy.
- **Python automation:** Automated the NV-orientation measurement protocol using Python.

Undergraduate Student Researcher • Indian Institute of Technology Madras (IITM), Chennai

Advisor: Prof. Basudev Roy May 2018 - Dec 2018

- **Optical tweezers & data acquisition:** Built and integrated a quadrant-photodiode-based detection and data acquisition system for a microscope-based optical tweezer platform, enabling scattered intensity measurements.
- **LabVIEW instrumentation:** Built a LabVIEW interface for automated data acquisition, storage, and analysis.

Education

Ph.D. Candidate, Physics • University of California, Santa Barbara • 2020 - Dec 2026 [Expected]

M.A., Physics • University of California, Santa Barbara • 2020 - 2023

Bachelor of Technology • Indian Institute of Technology Roorkee • 2016 - 2020
Major in Engineering Physics with Minor in Applied Mathematics

Technical Skills

Instrumentation: Hands-on experience with lasers, photodetectors, DAQs, oscilloscopes, spectrum analyzers, RF/microwave electronics, telecom and visible optical components (free-space and fiber-based), high-speed optical modulators and switches (AOMs, EOMs), and digital pulse-generation hardware.

Programming & automation: Python (NumPy, SciPy, Pandas, nidaqmx, Matplotlib, PyQt5, tkinter), instrument-control APIs, automated data acquisition, feedback control, measurement synchronization, Git, C++, LabVIEW.

Modeling & simulation: COMSOL finite-element simulation, MATLAB, quantitative modeling, parameter extraction, experimental fitting, and comparison of measured device performance with physical models.

Cryogenics: Dilution refrigerators, 4 K helium cryostats, cryogenic optical/RF/DC integration.

Electronics & hardware: Circuit design & debugging, soldering, PCB assembly, instrumentation interfacing, and analog/digital signal troubleshooting.

Machining skills: Milling, lathe operation, drilling.

Publications

V. Dharod et al., "**Ground-state operation of diamond optomechanical crystals**", *Manuscript in preparation*

H. Oh†, **V. Dharod**†, C. Padgett†, et al., "**Spin-embedded diamond optomechanical resonator with a mechanical quality factor exceeding one million**", *Optica* **13**, 485–490 (2026).
[press coverage] †Co-first authors.

S. Malik, F. M. Mayor, W. Jiang, H. Oh, C. Padgett, **V. Dharod**, J. Venkatraman, A. C. B. Jayich, A. Safavi-Naeini, "**Integrated phononic waveguide on thin-film lithium niobate on diamond**", *Applied Physics Letters* **127**, 154001 (2025).

M. Koppenhöfer, C. Padgett, J. V. Cady, **V. Dharod**, H. Oh, A. C. B. Jayich, and A. A. Clerk, "**Single-spin readout and quantum sensing using optomechanically induced transparency**", *Physical Review Letters* **130**, 093603 (2023).

D. Bhatt, R. Vaippully, B. Kharbanda, A. D. Ranjan, R. Sulochana, **V. Dharod**, D. Satapathy, B. Roy, "**Detection of self-generated nanowaves on the interface of an evaporating sessile water droplet**", *Optics Express* **27**, 31900–31912 (2019).

Honors & Awards

Quantum Foundry Associateship	Dec 2020 - 2026
IIT Roorkee Heritage Foundation Excellence Award	Mar 2020
ThinkSwiss Research Scholarship Award	Feb 2020
Chennupati and Vidya Jagadish Endowment Scholarship Award	Jan 2019
National Graduate Physics Examination Gold Medal (NGPE 2018)	June 2018